

DIGISIM PS-1

Team Name- Kaizer

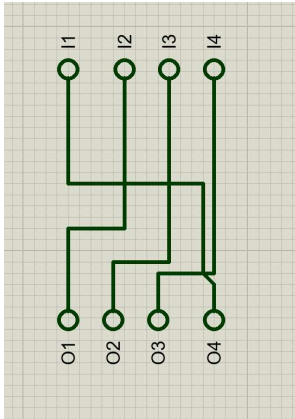
Member 1- Pratyush Meshram

Member 2- Aditya Kishore

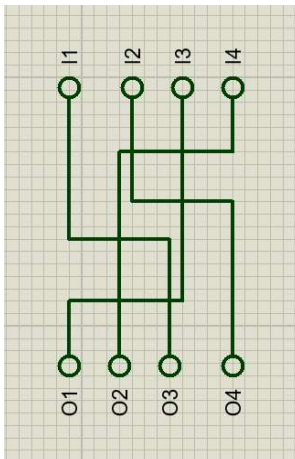
Combinational Circuit approach (Cost= Rs. 8)-

Round key generation:

- For 1 bit circular shift, we interchanged the input wires to the output

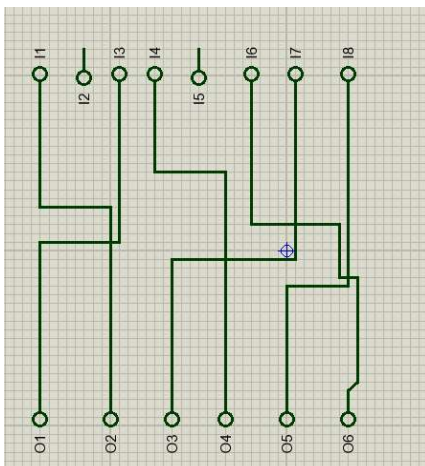


- For 2 bit circular shift



- Now, the left half and the right half of 1 (or 2) bit circular shift is merged and compressed by the compressor block which is present in the child sheet of the round key block:

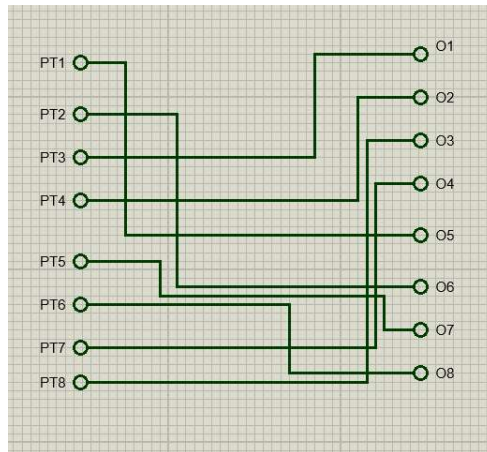
Internal wiring of compressor block:



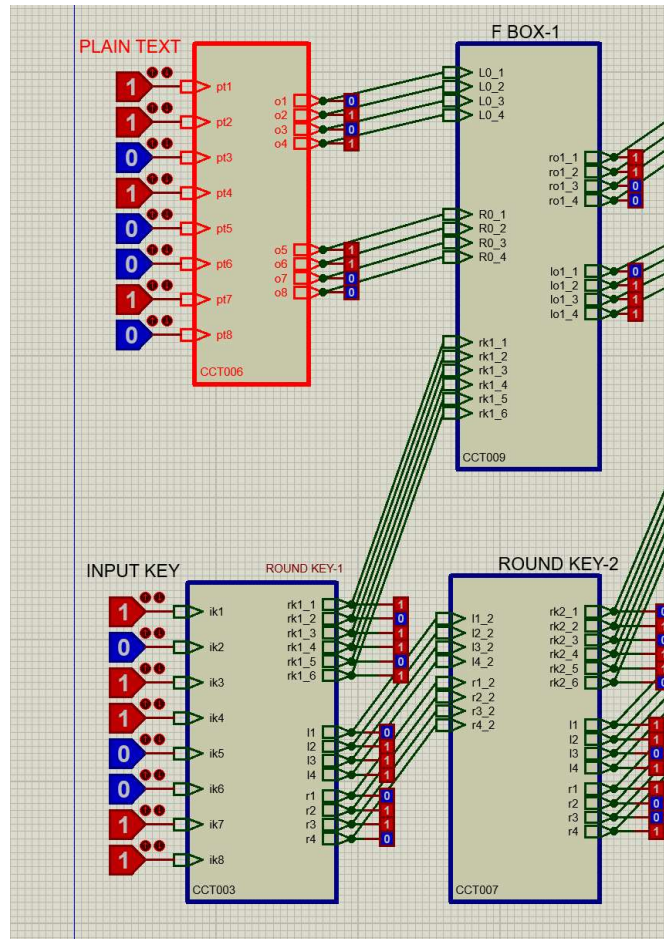
The outputs (O1-6) is the required round key of that round

Function (f) generator:

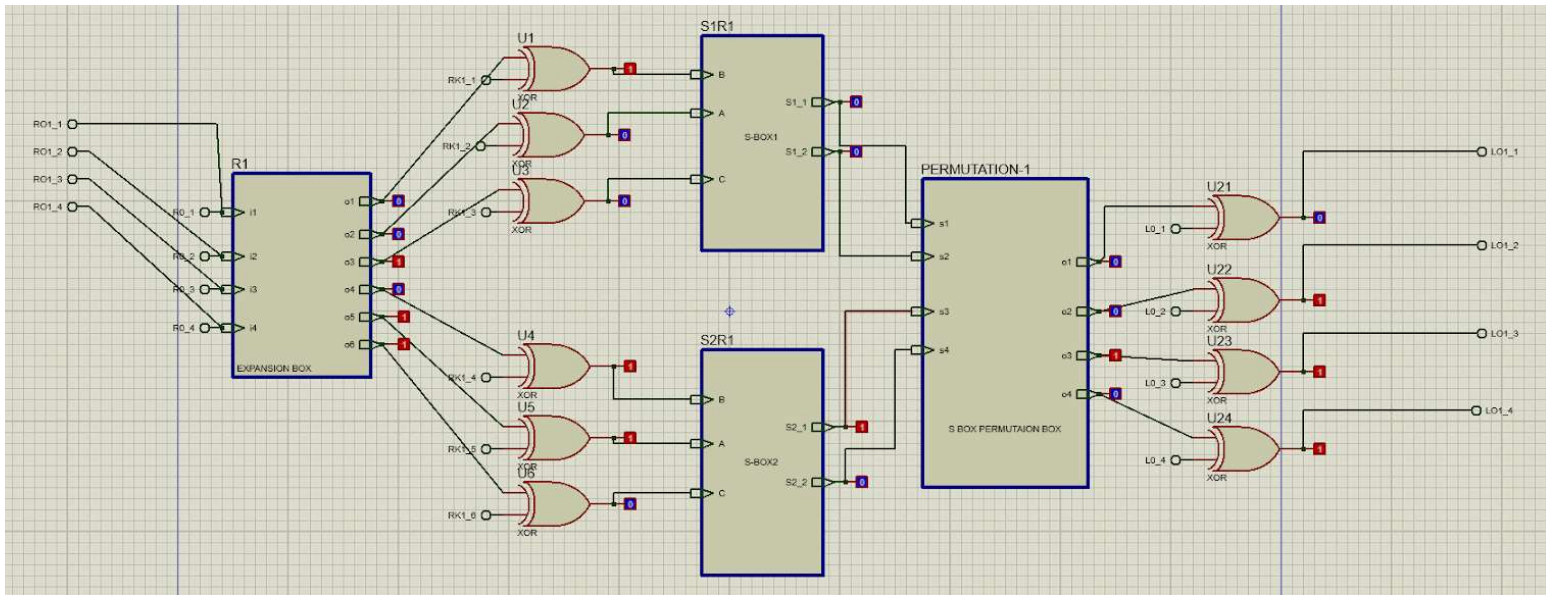
- The input plain text is permuted but interchanging the input wires to the output:



- Now this output is sent to the F box, where function f is generated and is xored with the left half of the permuted plain text



Internal circuit of the F box:

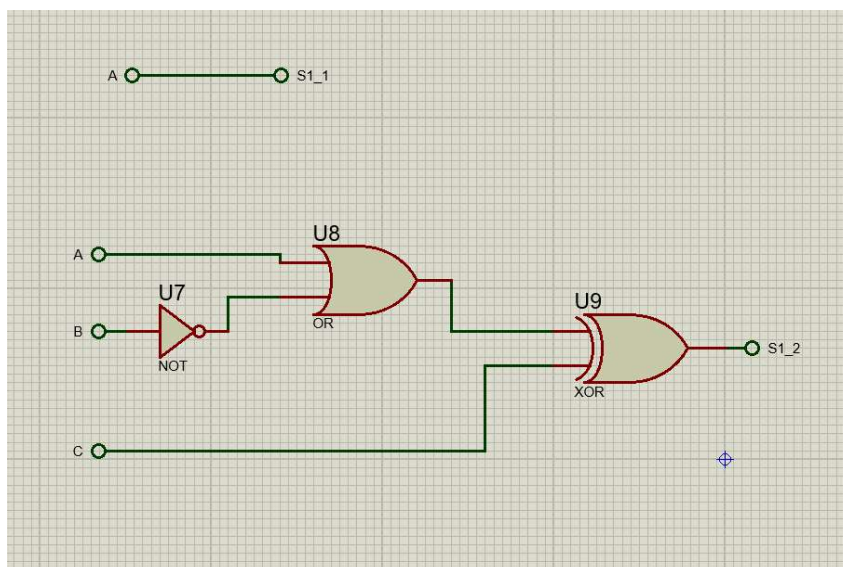


- There's an expansion box, where the 4 bit right half of the permuted text is expanded to 6 bits.
- The first 3 bits are xored with first 3 bits of round key and sent as input to S-Box 1, and the other 3 bits are xored with other 3 bits of round key and sent as input to S-Box 2

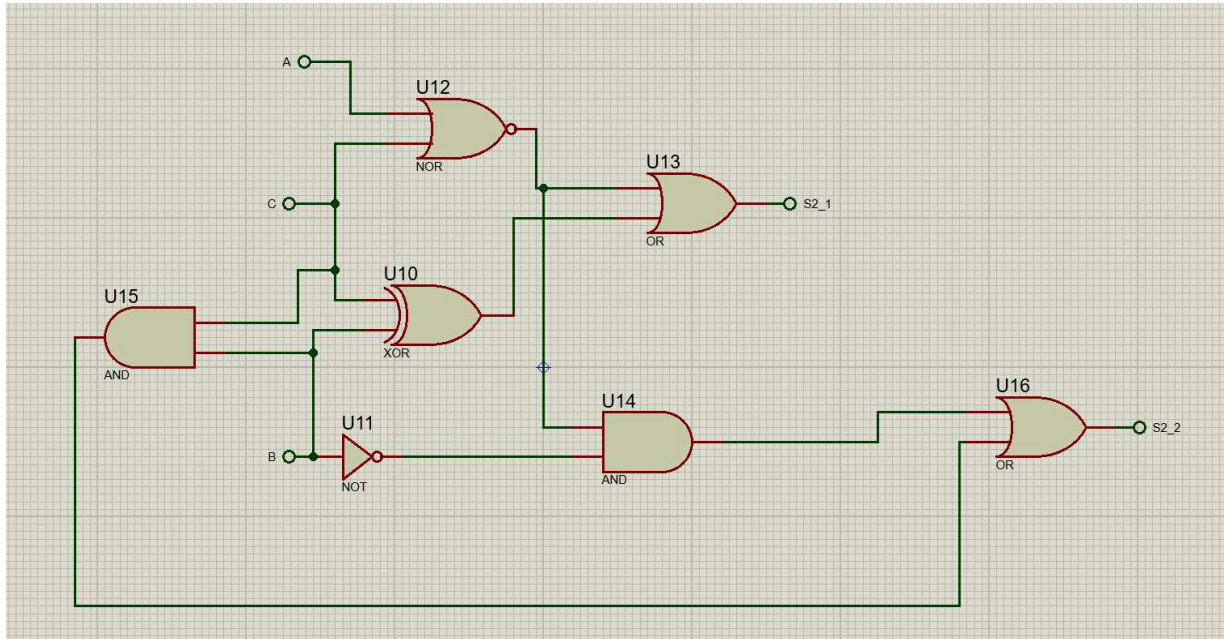
S-Box 1:

For the given outputs of S boxes in the PS, we made k-map for the first and second bit of S boxes output

Internal structure of S-box 1:

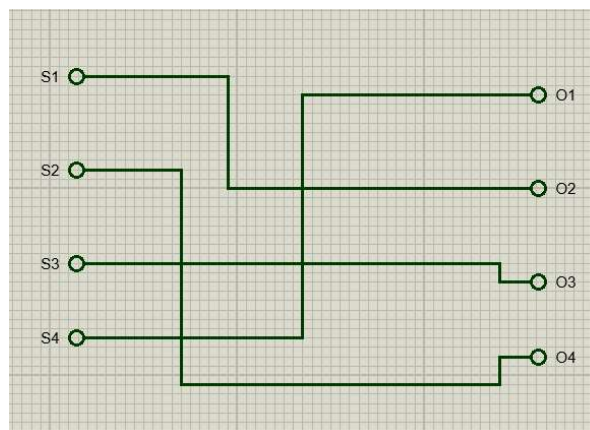


Internal structure of S-box 2:

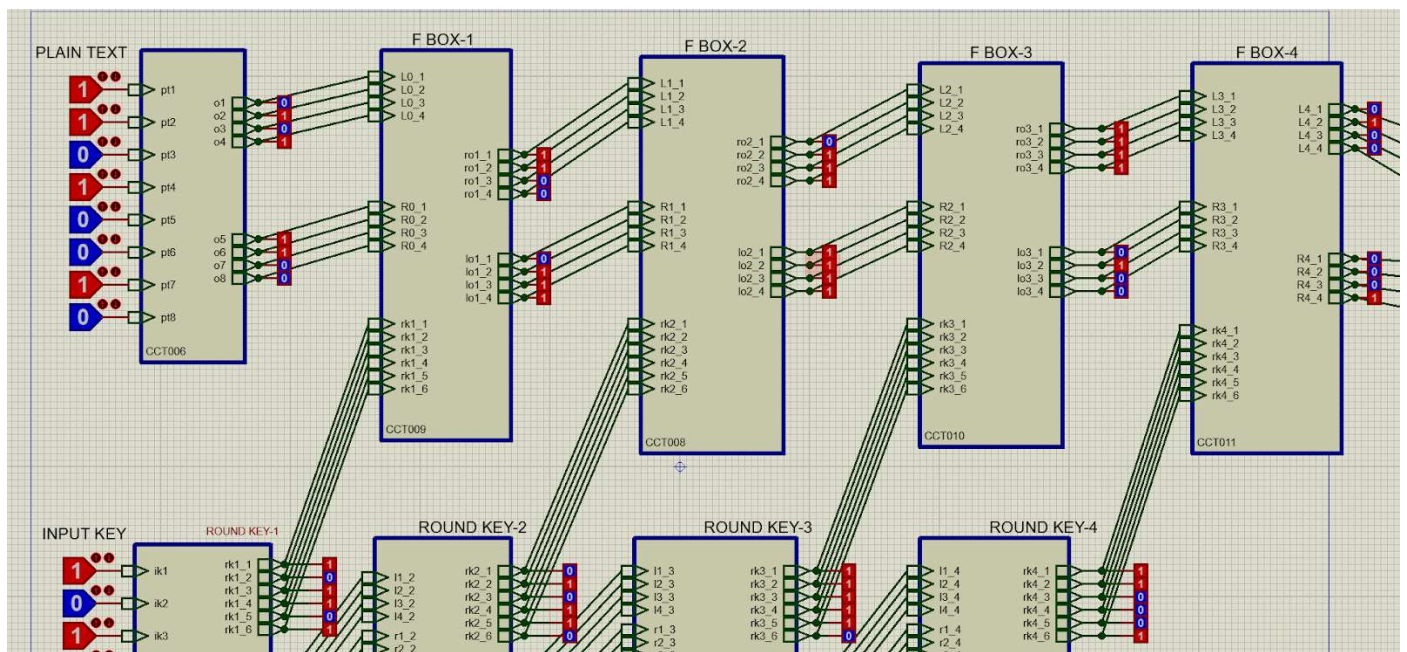


The output, S_{n_1} , S_{n_2} ($n=1,2$) represents the first and second bit of output respectively

- Now the output of the S boxes is send to the S-box permutation block
Internal structure if it:

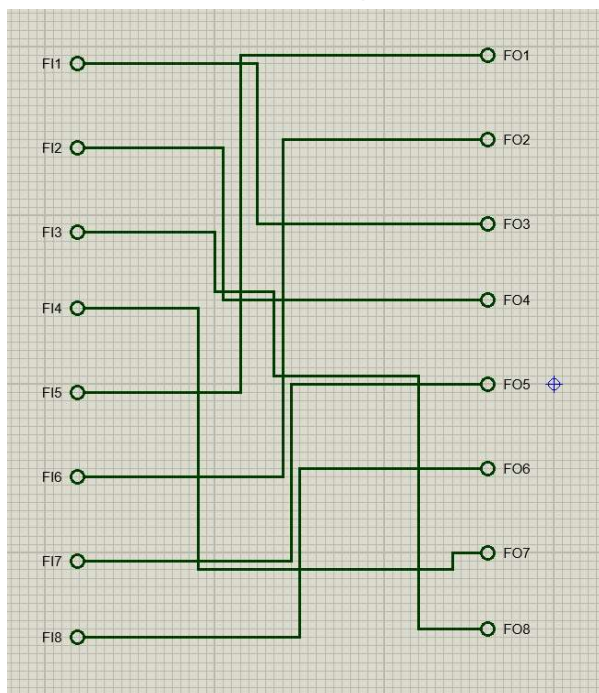


- The output of S-box permutation block is xored with the left half of the permuted plain text to give the output R_1
- $L_1 = R_0$
- We, now have to repeat this and give the left half output of the previous round as the input of the right half of the current round:



- After round 4, the output (L4,R4) is sent to the Inverse permutation block

Internal structure of Inverse permutation block:



The output of this block gives us the encrypted message.

Cost Calculation:

- 10 gates in F-box:
Cost= $10 \times 0.1 = \text{Rs. } 1$
- 3 gates in S-box 1:
Cost= $3 \times 0.1 = \text{Rs. } 0.3$
- 7 gates in S-box 2:
Cost= $7 \times 0.1 = \text{Rs. } 0.7$

Total cost of 1 round= $1+0.3+0.7 = \text{Rs. } 2$

Total cost of circuit= $4 \times 2 = \text{Rs. } 8$